

LAB RESOURCE UTILIZATION PLATFORM

MILESTONE 2 – IMPLEMENTATION & OUTPUT DOCUMENTATION

Team-3

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1. Project Overview

The Lab Resource Utilization Platform is a full-stack application designed to help research institutions, universities, and laboratories manage laboratory equipment, schedule resource access, share expensive equipment between institutions, monitor utilization, manage maintenance, and analyze resource efficiency.

The platform follows a role-based access model so that users can access functionality appropriate to their assigned role.

2. Milestone 2 Overview

Milestone 2 focuses on implementing and demonstrating an integrated resource-utilization workflow rather than developing disconnected screens. The milestone covers real-time equipment utilization, heatmap and idle-time detection, inter-institution resource sharing, external booking and access, utilization and demand analysis, and waitlist and booking optimization.

3. Milestone 2 Objectives

- Implement real-time equipment utilization monitoring.
- Provide utilization heatmap and idle-time detection.
- Support inter-institution equipment/resource sharing.
- Support external booking and access workflows.
- Provide utilization-rate and demand analysis.
- Provide waitlist and booking optimization.
- Connect frontend functionality with APIs, backend services, and the database.
- Enforce role-based access and prevent unauthorized functionality.

4. User Roles

Role	Main Responsibility	Milestone 2 Relevance
Researcher / Student	Search, check availability, book equipment, manage bookings and waitlists	External booking, availability, waitlist
Lab Technician	Equipment and maintenance-related tasks	Equipment operational status / maintenance
Lab Manager	Manage resources and monitor utilization	Utilization, heatmap, demand, sharing, approvals
Department Head	Department-level resource monitoring	Utilization, heatmap, demand, sharing
Institution Administrator	Organization-wide resource intelligence	Resource sharing, analytics, organization-level data
System Administrator	Users, roles, access and system monitoring	Role-based access and security

5. Milestone 2 Role Mapping

Milestone 2 Task	Primary Role	Supporting Role
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1. Real-time Equipment Utilization	Lab Manager / Department Head	Researcher
2. Heatmap & Idle Time Detection	Lab Manager / Department Head	—
3. Inter-Institution Resource Sharing	Lab Manager / Department Head	Institution Administrator
4. External Booking & Access	Researcher / Student	Lab Manager
5. Utilization Rate & Demand Analysis	Lab Manager / Department Head	Institution Administrator
6. Waitlist & Booking Optimization	Researcher / Student	Lab Manager

6. Implementation & Output Screenshots

6.1 Role-Based Login and Access

The application provides role-based login and loads the appropriate dashboard and navigation according to the authenticated user's role. Unauthorized functionality should not be accessible to users outside their permitted role.

Figure 6.1: Role-Based Login / Authentication

← → ↻ localhost:5173/register ☆ 📁 📄 m

RU

JOIN RESOURCEHUB

**One workspace
for every
resource.**

Create your account and make institutional
equipment easier to discover, book and

GET STARTED

Create your account

Use your institutional details to set up your profile.

First name

Last name

Email address

Researcher / Student
Lab Technician
Lab Manager
Department Head
Institution Administrator
System Administrator

Researcher / Student ▼

6.2 Researcher / Student Dashboard

The Researcher / Student workflow allows users to find laboratory resources, check equipment availability, make bookings, view booking history, and manage waitlist status.

- View and search available equipment.
- Check equipment status and availability.
- Book available equipment.
- View upcoming, current, completed, and cancelled bookings.
- Join a waitlist when equipment is unavailable.
- View waitlist status and notifications.

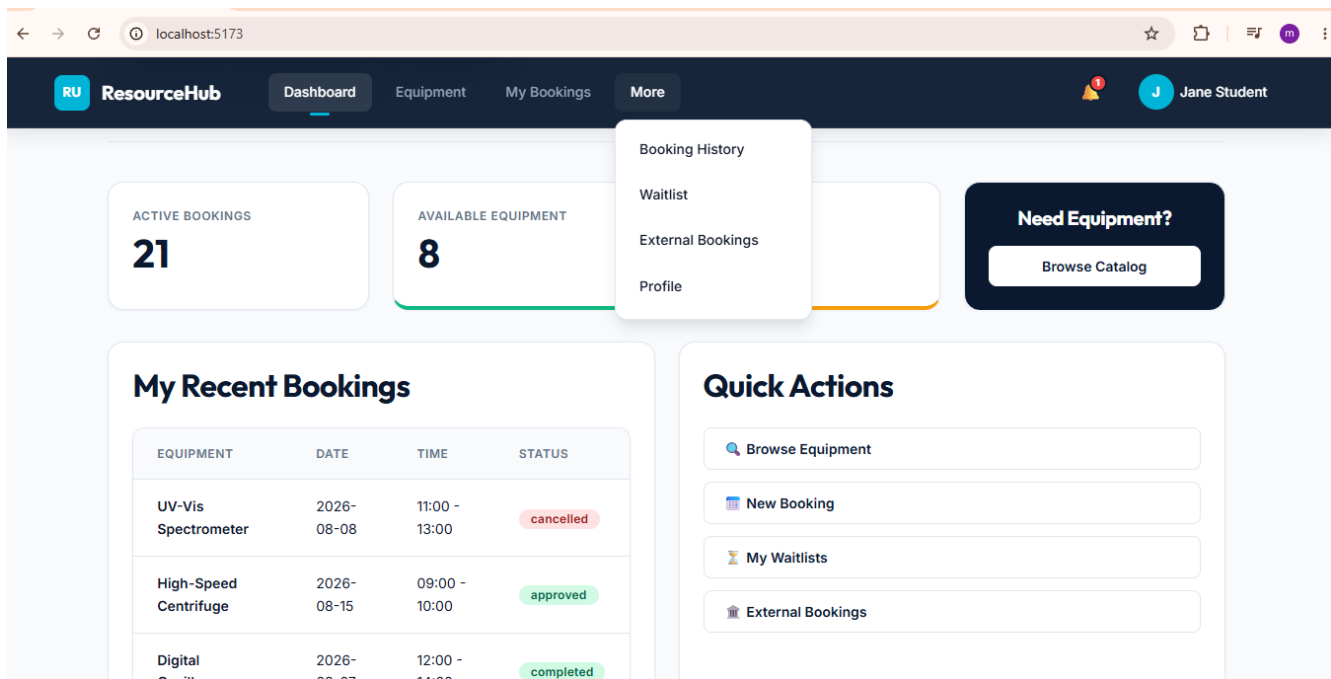


Figure 6.3: Researcher / Student Dashboard

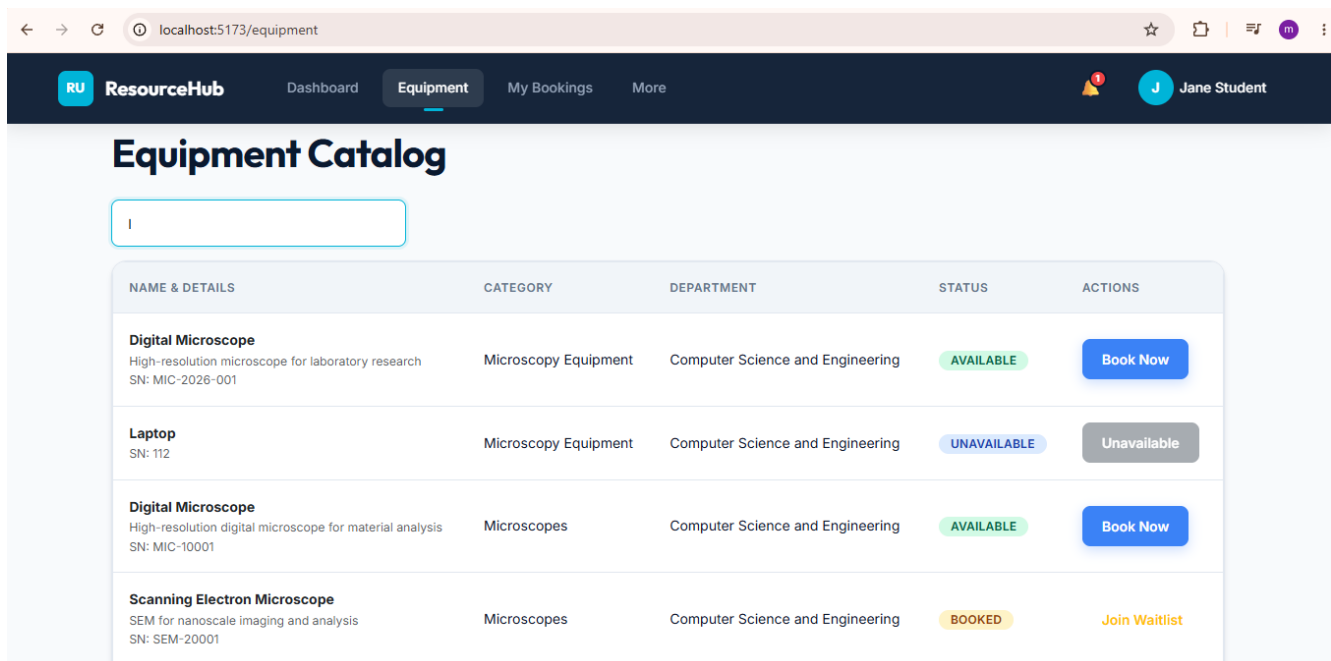


Figure 6.4: Equipment Search / Availability

The screenshot shows a web browser window with the URL `localhost:5173/external-bookings`. The page header includes the 'RU ResourceHub' logo and navigation links: 'Dashboard', 'Equipment', 'My Bookings', and 'More'. A user profile for 'Jane Student' is visible in the top right corner. The main content area is titled 'External Bookings' and contains a form titled 'Request External Equipment Access'. The form has four input fields: 'Equipment' (containing '3D Printer (DY Patil International University)'), 'Requesting Institution' (containing 'gec'), 'Start Time' (containing '13-08-2026 21:06'), and 'End Time' (containing '14-08-2026 21:06'). There are calendar icons next to the time fields. A 'Purpose' dropdown menu is set to 'research'. A blue 'Submit Request' button is located below the form. At the bottom of the page, there is a filter section with 'All Statuses' selected, a 'Filter' button, and a 'Clear' button.

Figure 6.5: Equipment Booking

6.3 Real-Time Equipment Utilization

This functionality monitors equipment operational status and utilization levels. The utilization workflow records usage data and updates the utilization view so that laboratory managers can identify how resources are being used.

- View current equipment status.
- View utilization percentage and usage trends.
- Identify idle equipment.
- Identify high-, medium-, and low-utilization patterns.
- Monitor peak usage patterns.

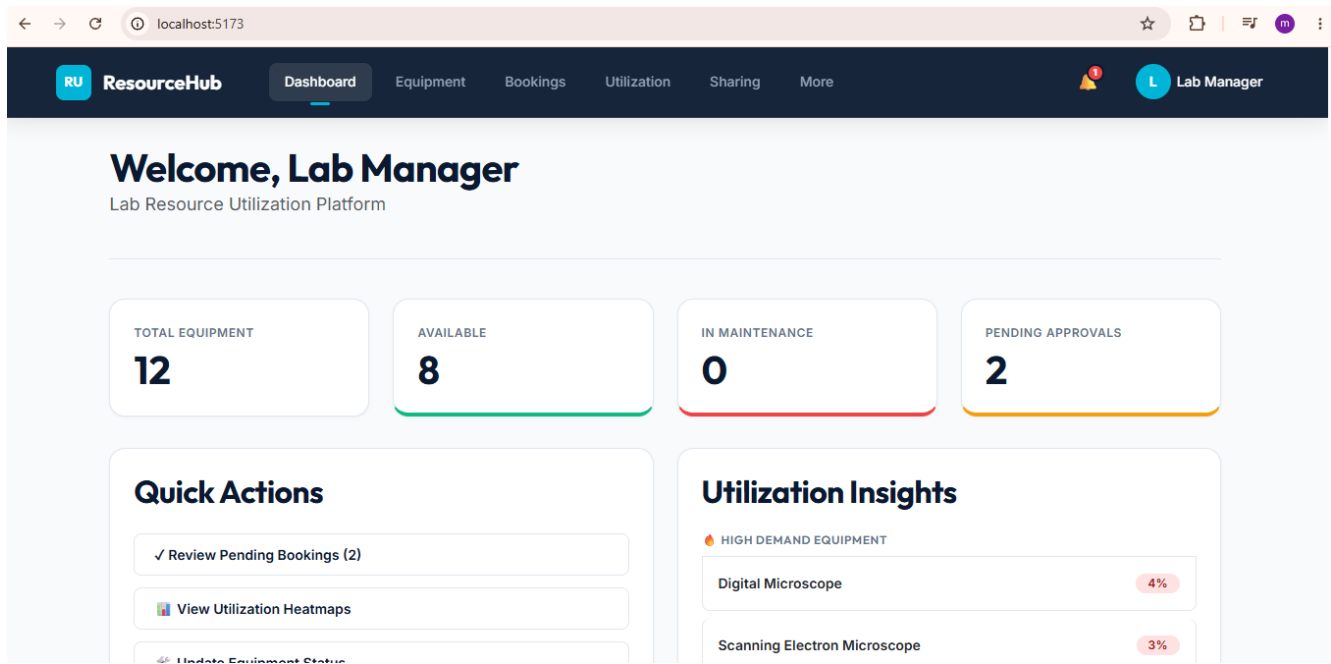


Figure 6.6: Real-Time Equipment Utilization

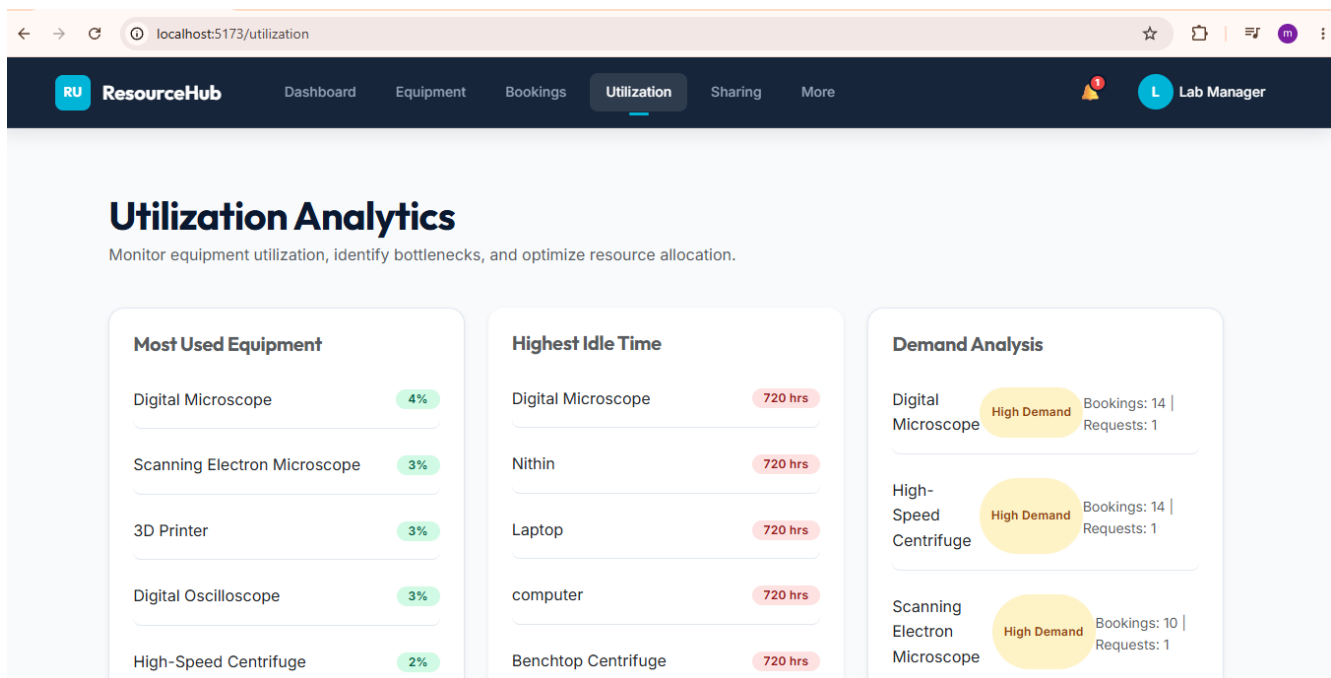


Figure 6.7: Equipment Status Change

Screenshot to insert: Insert screenshot showing Available / Booked / In Use / Maintenance or another implemented state.

6.4 Utilization Heatmap and Idle Time Detection

The heatmap provides a visual representation of equipment usage patterns. It is used to identify peak usage periods and idle resources and supports resource-allocation decisions by laboratory managers and department heads.

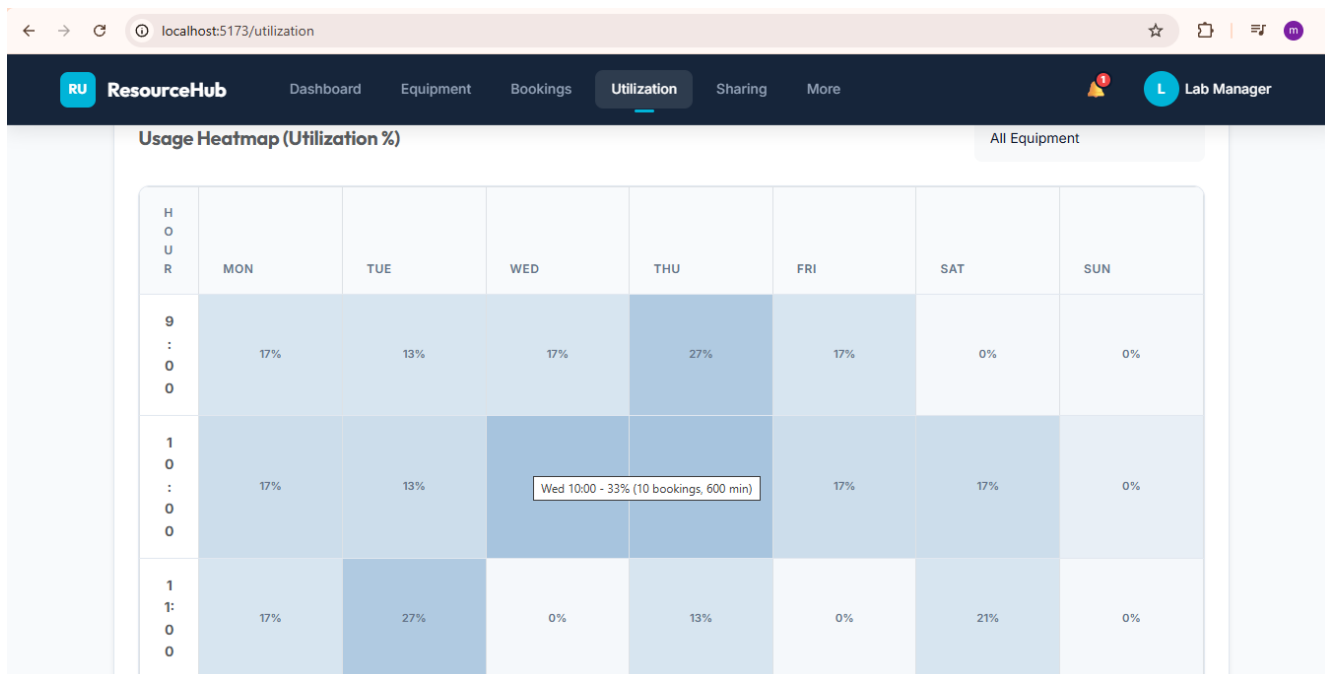


Figure 6.8: Utilization Heatmap

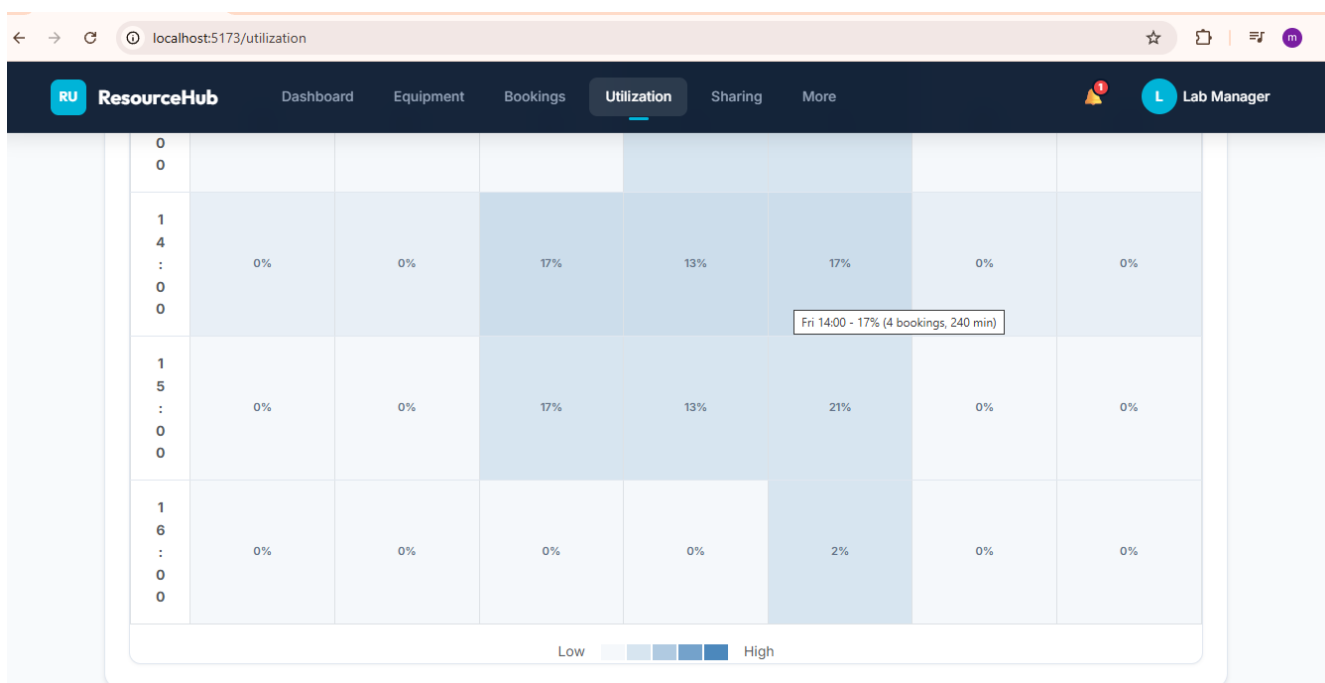


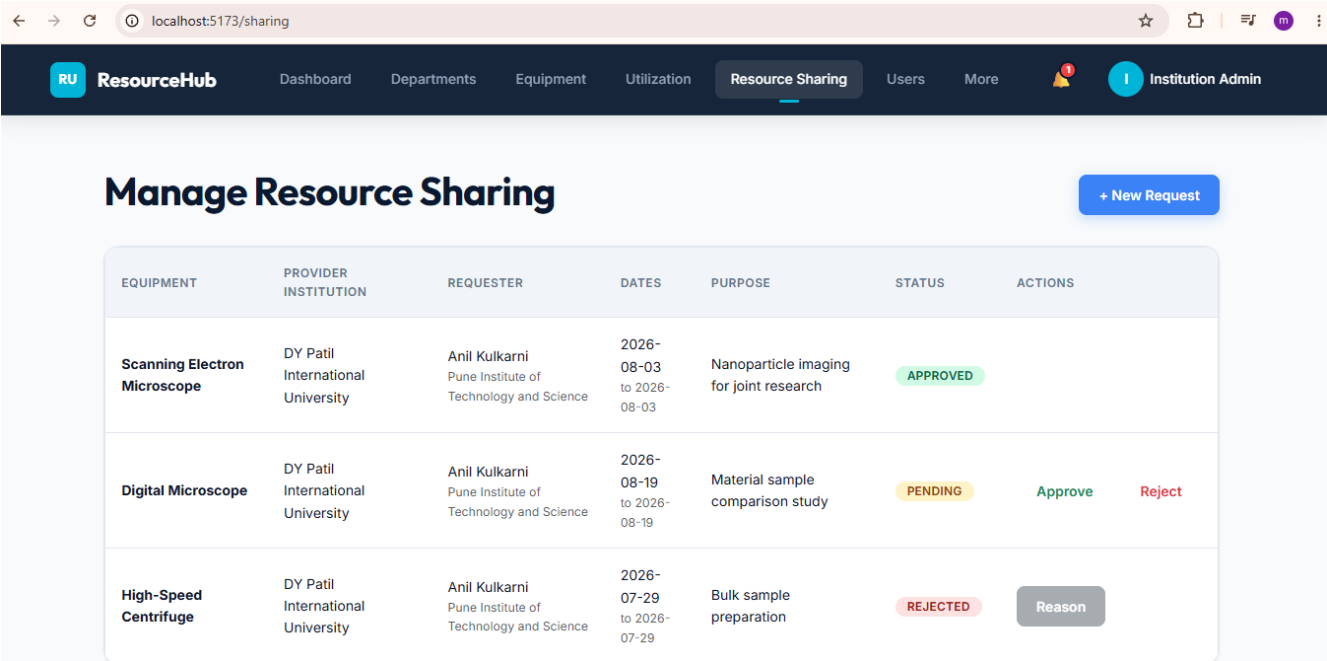
Figure 6.9: Idle Equipment Detection

6.5 Inter-Institution Resource Sharing

The platform supports sharing of laboratory equipment between institutions. Authorized users can view shared equipment, create or review sharing/access requests, and manage approvals or rejections.

- View shared equipment.
- View incoming sharing/access requests.
- Approve or reject sharing requests.

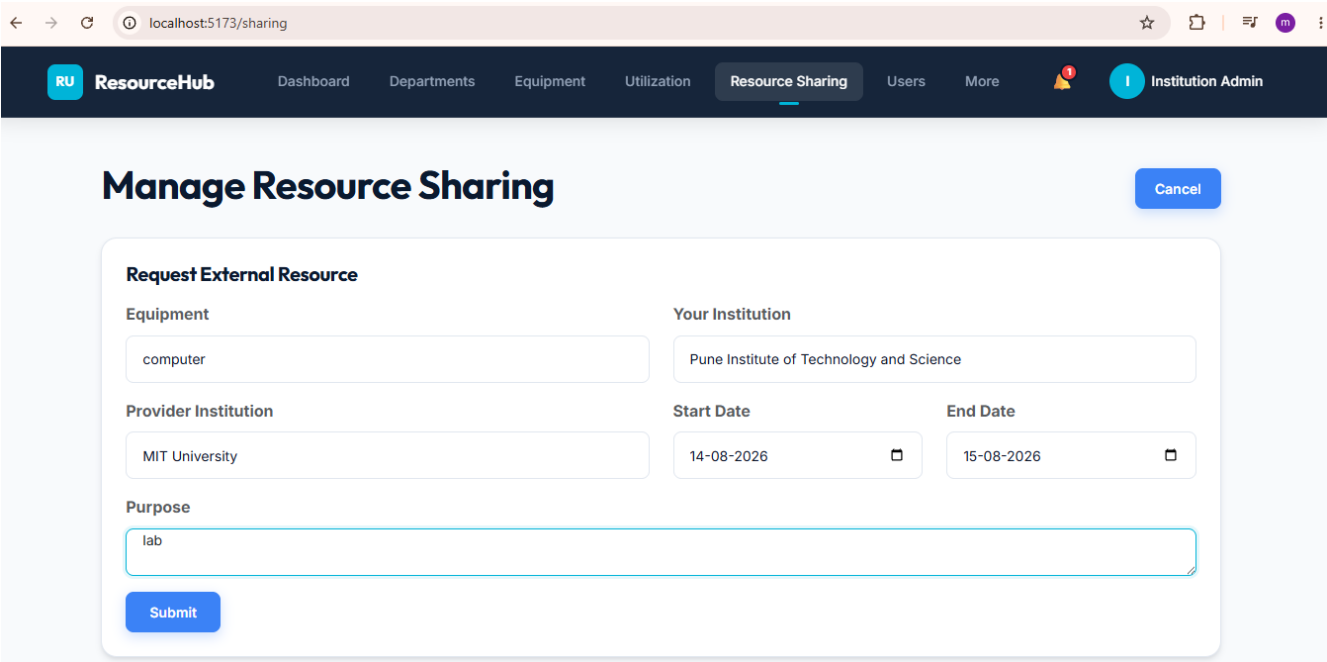
- Monitor shared-equipment scheduling.



The screenshot shows the 'Manage Resource Sharing' interface. At the top, there's a navigation bar with 'RU ResourceHub' and links to Dashboard, Departments, Equipment, Utilization, Resource Sharing (active), Users, and More. A user profile 'Institution Admin' is logged in. Below the navigation bar, the title 'Manage Resource Sharing' is displayed with a '+ New Request' button. The main content is a table with the following data:

EQUIPMENT	PROVIDER INSTITUTION	REQUESTER	DATES	PURPOSE	STATUS	ACTIONS
Scanning Electron Microscope	DY Patil International University	Anil Kulkarni Pune Institute of Technology and Science	2026-08-03 to 2026-08-03	Nanoparticle imaging for joint research	APPROVED	
Digital Microscope	DY Patil International University	Anil Kulkarni Pune Institute of Technology and Science	2026-08-19 to 2026-08-19	Material sample comparison study	PENDING	Approve Reject
High-Speed Centrifuge	DY Patil International University	Anil Kulkarni Pune Institute of Technology and Science	2026-07-29 to 2026-07-29	Bulk sample preparation	REJECTED	Reason

Figure 6.10: Resource Sharing



The screenshot shows the 'Request External Resource' form. The title 'Manage Resource Sharing' is at the top with a 'Cancel' button. The form is titled 'Request External Resource' and contains the following fields:

- Equipment:** A text input field containing 'computer'.
- Your Institution:** A text input field containing 'Pune Institute of Technology and Science'.
- Provider Institution:** A text input field containing 'MIT University'.
- Start Date:** A date input field containing '14-08-2026' with a calendar icon.
- End Date:** A date input field containing '15-08-2026' with a calendar icon.
- Purpose:** A text input field containing 'lab'.
- Submit:** A blue button at the bottom left of the form.

Figure 6.11: Sharing Request

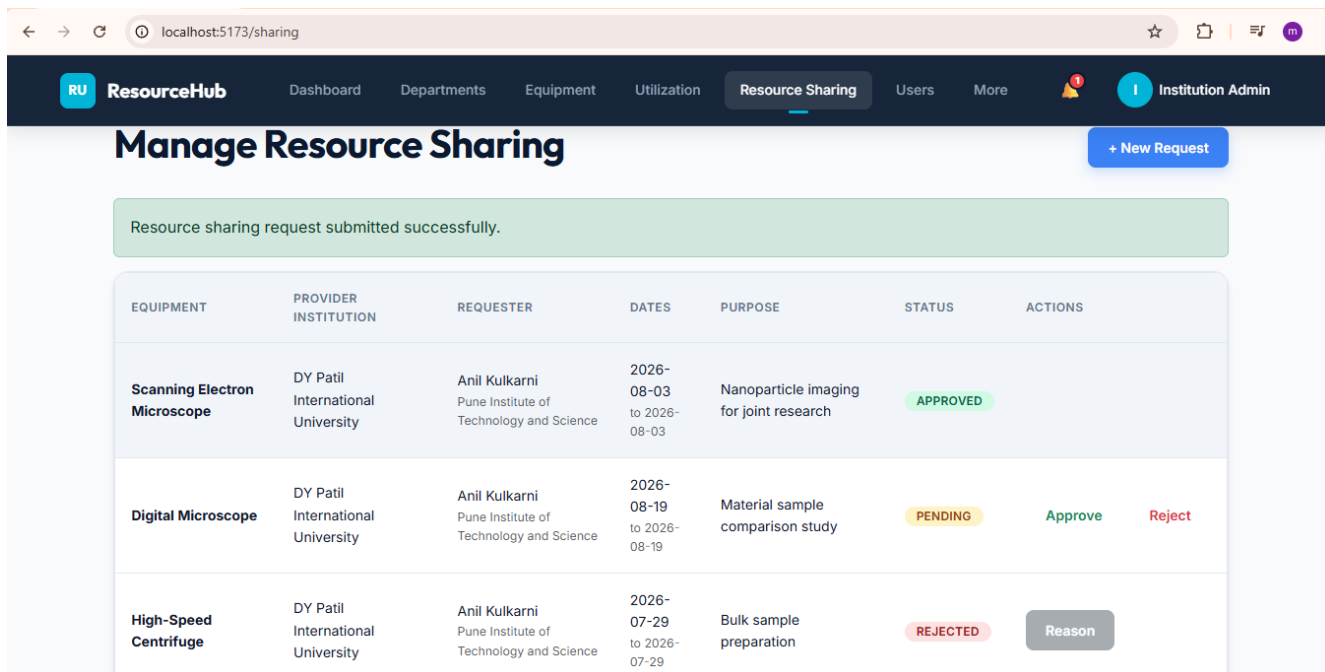


Figure 6.12: Sharing Approval / Rejection

Screenshot to insert: Insert approval/rejection output screenshot here.

6.6 External Booking and Access

The external booking workflow allows a researcher or student to request access to available equipment while the Lab Manager supports the access-management process.

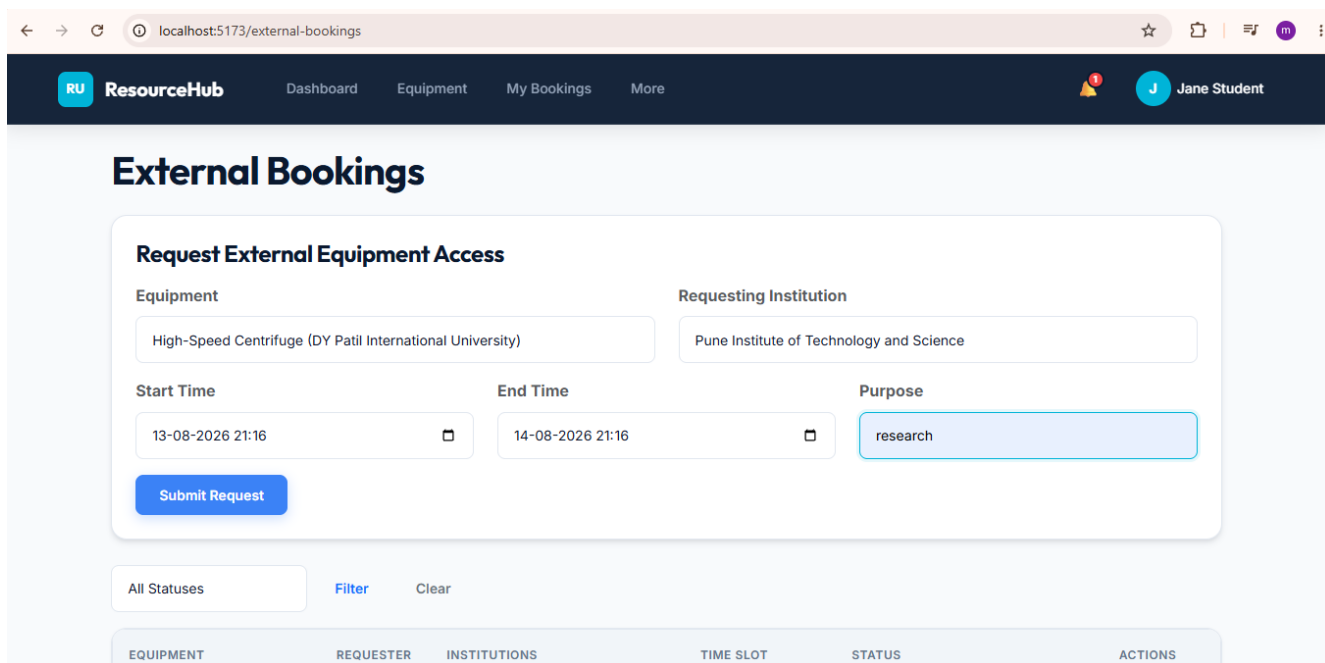


Figure 6.13: External Booking / Access Request

All Statuses		Filter	Clear		
EQUIPMENT	REQUESTER	INSTITUTIONS	TIME SLOT	STATUS	ACTIONS
Digital Microscope External sample imaging engagement	Anil Kulkarni	From: Pune Institute of Technology and Science To: DY Patil International University	30/07/2026, 10:00:00 to 30/07/2026, 13:00:00	Access: Session completed and samples returned	
Scanning Electron Microscope Nanocoating thickness verification	Anil Kulkarni	From: Pune Institute of Technology and Science To: DY Patil International University	20/08/2026, 09:00:00 to 20/08/2026, 13:00:00	APPROVED Access: Check in at Lab B reception, wear lab coat	Cancel

Figure 6.14: Booking Approval / Confirmation

6.7 Utilization Rate and Demand Analysis

Utilization and demand analysis provides information about equipment usage, high-demand resources, booking patterns, and utilization trends. The information can support decisions about resource allocation and equipment planning.

- Utilization percentage.
- Usage trends.
- Booking statistics.
- High-demand equipment.
- Peak usage patterns.
- Demand analysis.

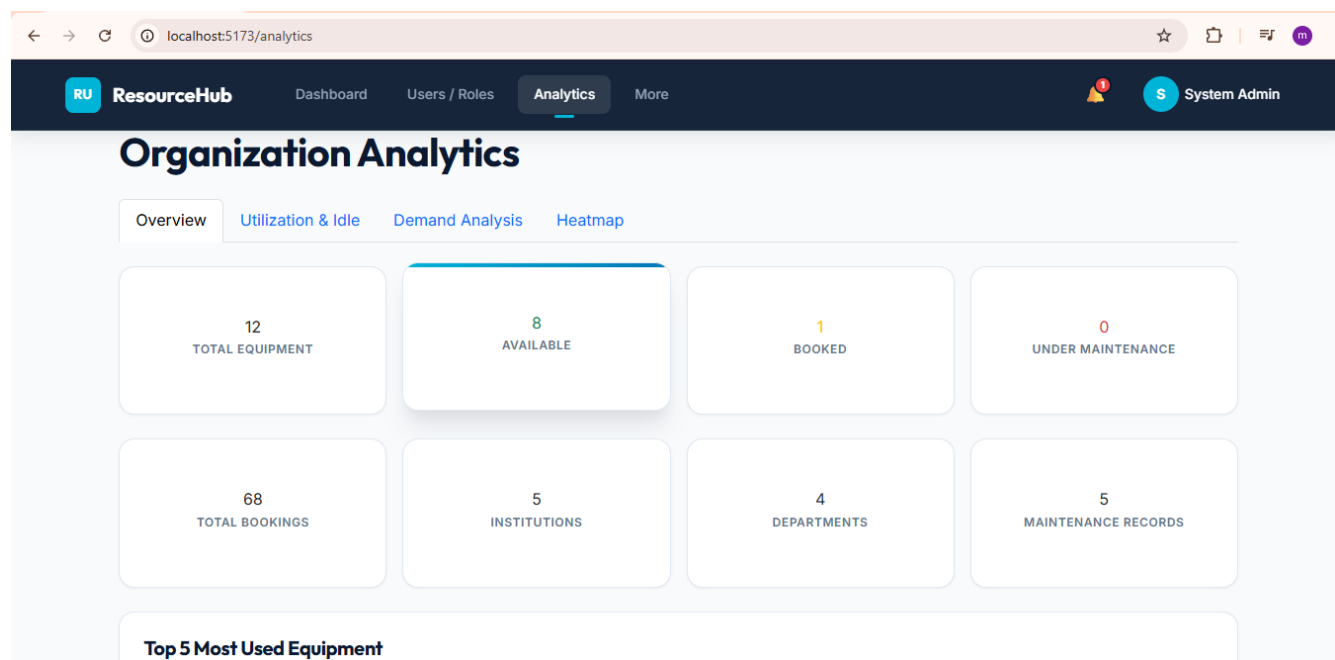


Figure 6.15: Utilization Analytics

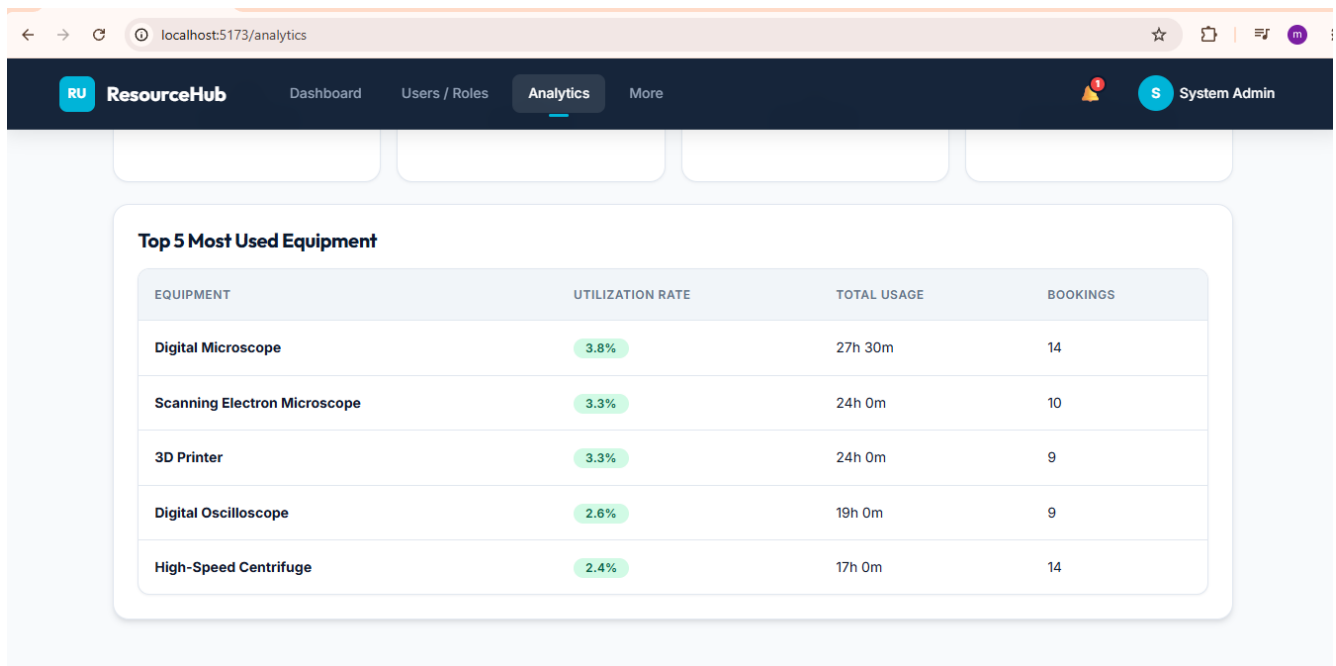


Figure 6.16: Demand Analysis / High-Demand Equipment

6.8 Waitlist and Booking Optimization

When equipment is unavailable, the researcher can join a waitlist. When the equipment becomes available, the waitlist and notification workflow can be triggered. Booking workflows should also prevent double booking of the same resource for overlapping availability.

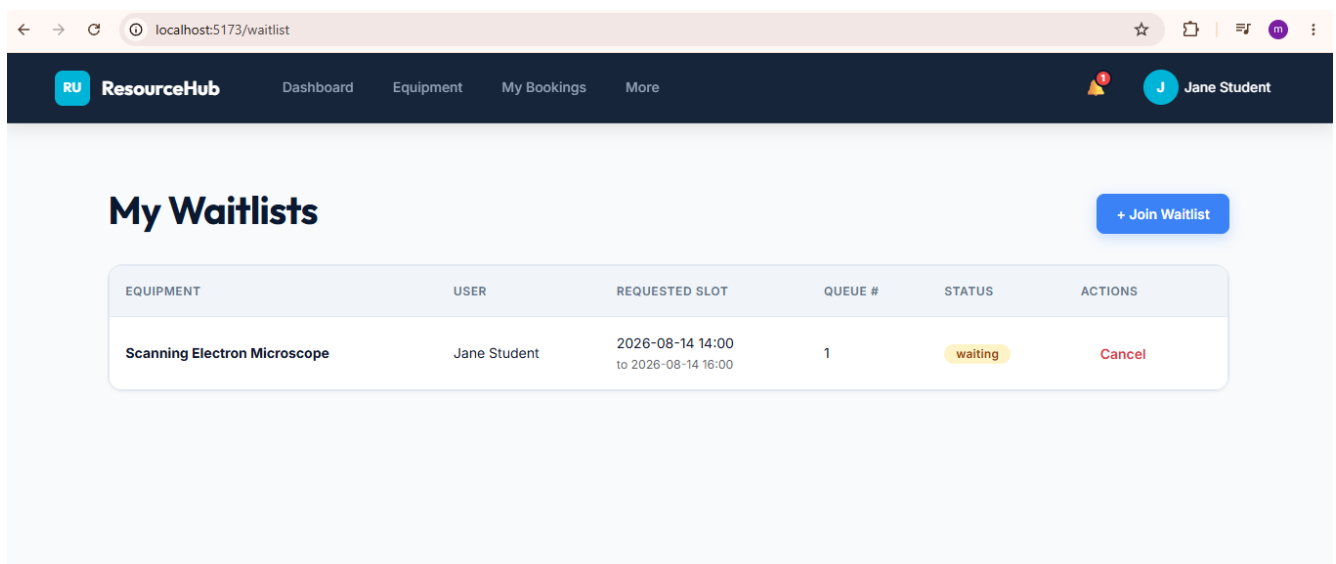


Figure 6.17: Unavailable Equipment / Waitlist Option

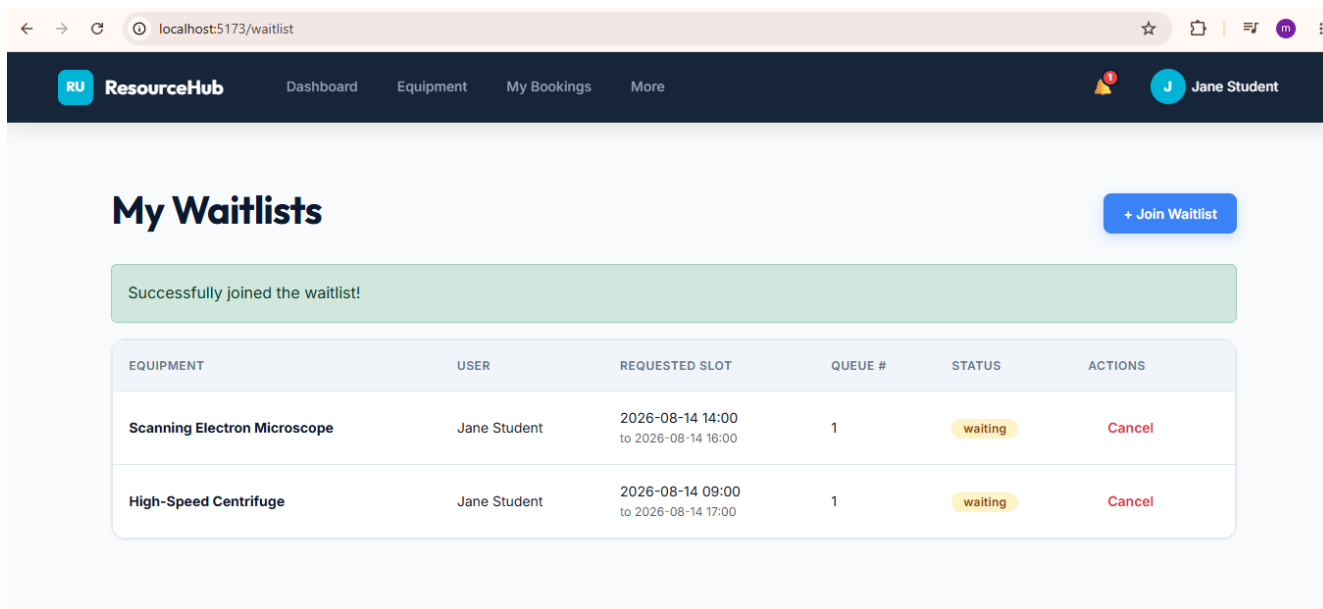


Figure 6.18: Waitlist Status

7. Lab Manager Dashboard

The Lab Manager dashboard is a key Milestone 2 interface for managing laboratory resources and monitoring utilization. It includes utilization, heatmap, booking statistics, no-show rates, maintenance overview, high-demand equipment, resource sharing, and sharing requests.

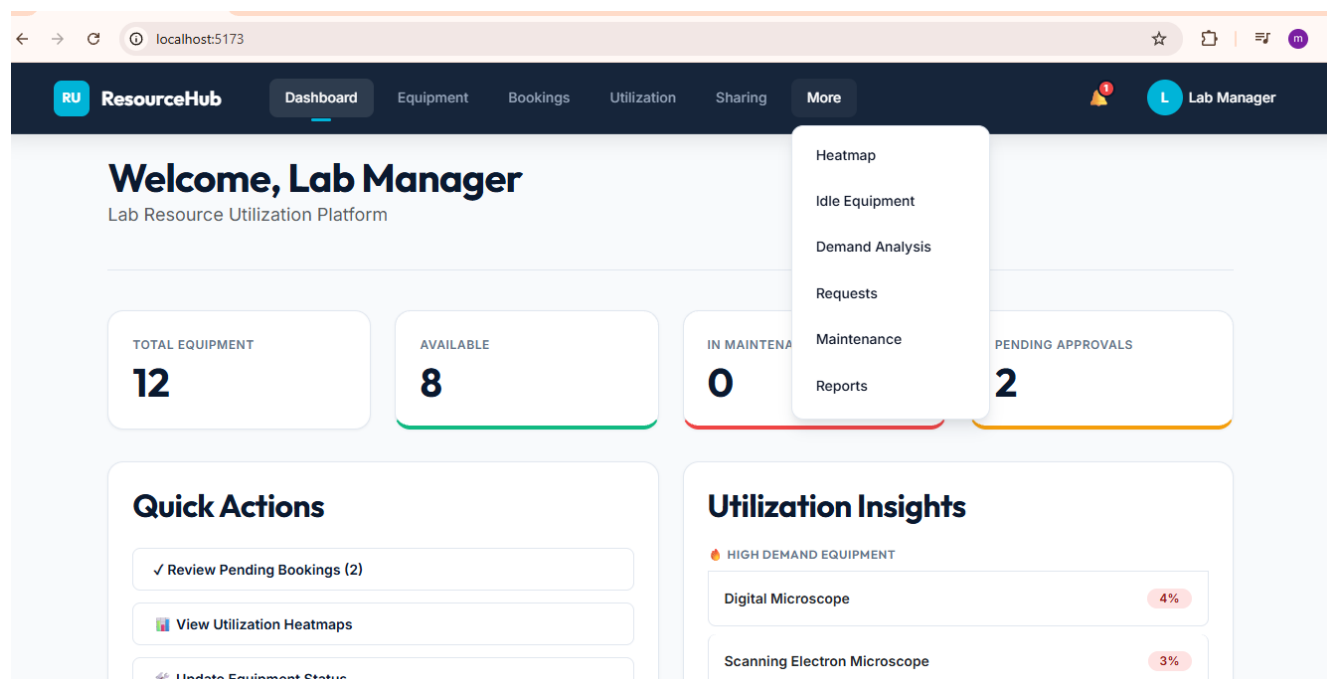


Figure 6.20: Lab Manager Dashboard

8. Department Head Dashboard

The Department Head can work with department-level information where the access model supports it. Relevant functionality includes department equipment, utilization, utilization heatmap, booking statistics, high-demand equipment, maintenance overview, sharing requests, and reports.

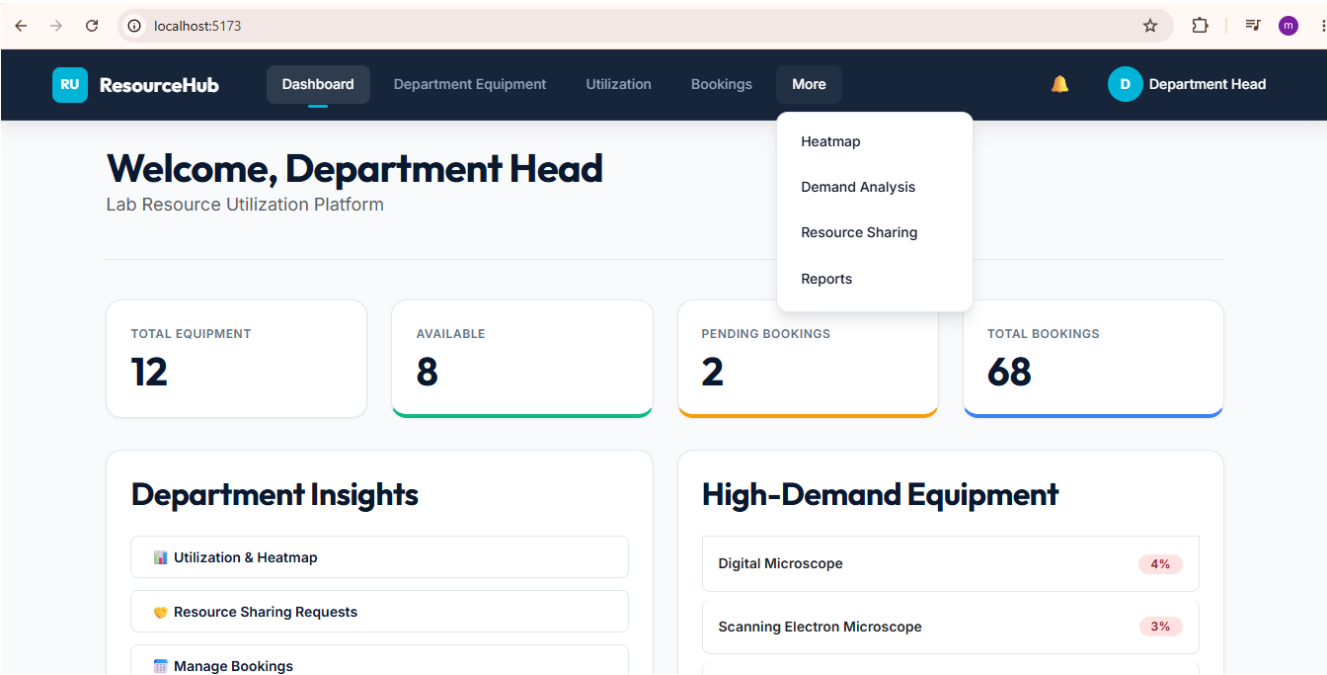


Figure 6.21: Department Head Dashboard

9. Institution Administrator Dashboard

The Institution Administrator provides organization-wide resource intelligence, including organization-wide utilization, cross-department resource sharing, procurement/cost analysis, equipment lifecycle information, user management, and reports.

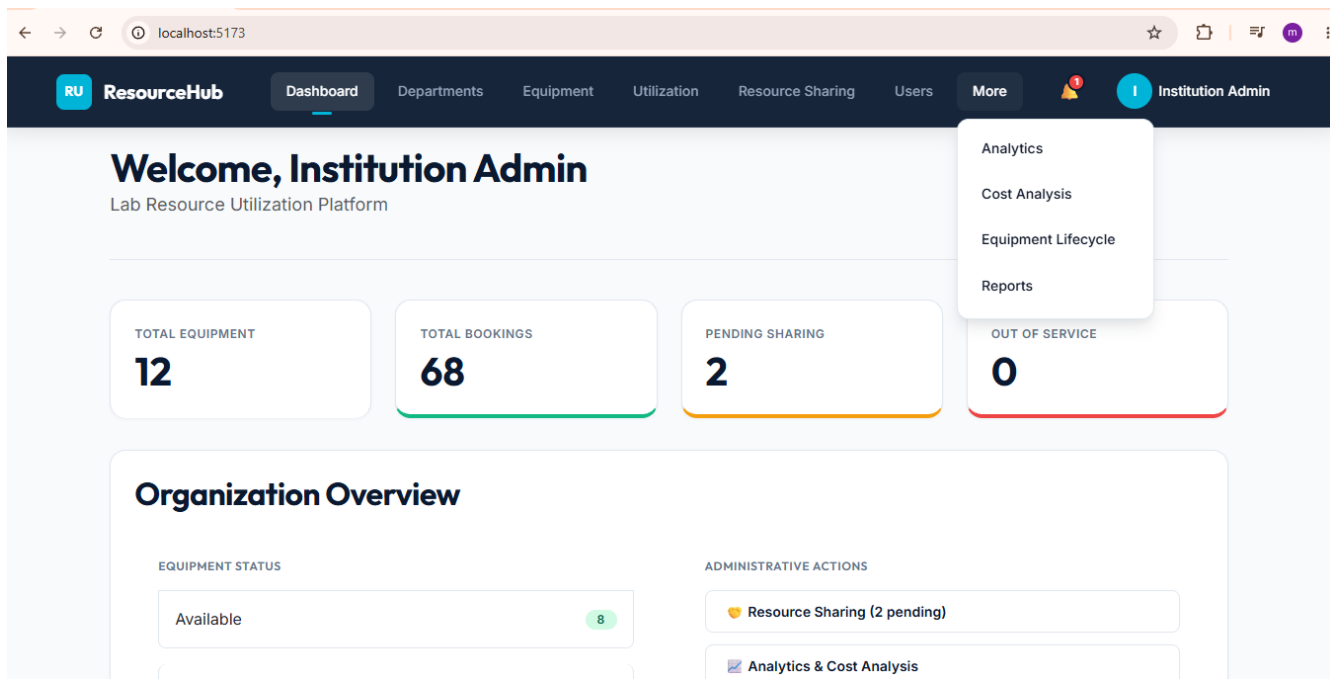


Figure 6.22: Institution Administrator Dashboard

10. End-to-End Milestone 2 Workflow

Milestone 2 is implemented as an integrated workflow. The expected sequence is:

1. Researcher searches for equipment.
2. Researcher checks equipment availability.
3. Researcher books available equipment.
4. Equipment status changes according to booking/usage state.
5. Usage data is recorded.
6. Utilization percentage is calculated.
7. Lab Manager dashboard and heatmap are updated.
8. If equipment is unavailable, the researcher can join the waitlist.
9. When equipment becomes available, the waitlist/notification workflow is triggered.
10. Institutions can share equipment and manage access requests.
11. Recommended/demand information supports resource planning.

localhost:5173/bookings

RU
ResourceHub

Dashboard
Equipment
My Bookings
More

Jane Student

My Bookings

New Booking Request

Equipment

7-Day Availability

TIM E	THU 13	FRI 14	SAT 15	SUN 16	MON 17	TUE 18	WED 19
09-10	●	●	●	●	●	●	●
10-11	●	●	●	●	●	●	●
11-12	●	●	●	●	●	●	●
12-	●	●	●	●	●	●	●

localhost:5173/bookings

RU
ResourceHub

Dashboard
Equipment
My Bookings
More

Jane Student

12-13	●	●	●	●	●	●	●
13-14	●	●	●	●	●	●	●
14-15	●	●	●	●	●	●	●
15-16	●	●	●	●	●	●	●
16-17	●	●	●	●	●	●	●

● Available
● Booked
● Pending
● Outside Hours

Selected Slot: 2026-08-15 | 09:00 to 10:00
AVAILABLE

Get Suggestions
Book Now

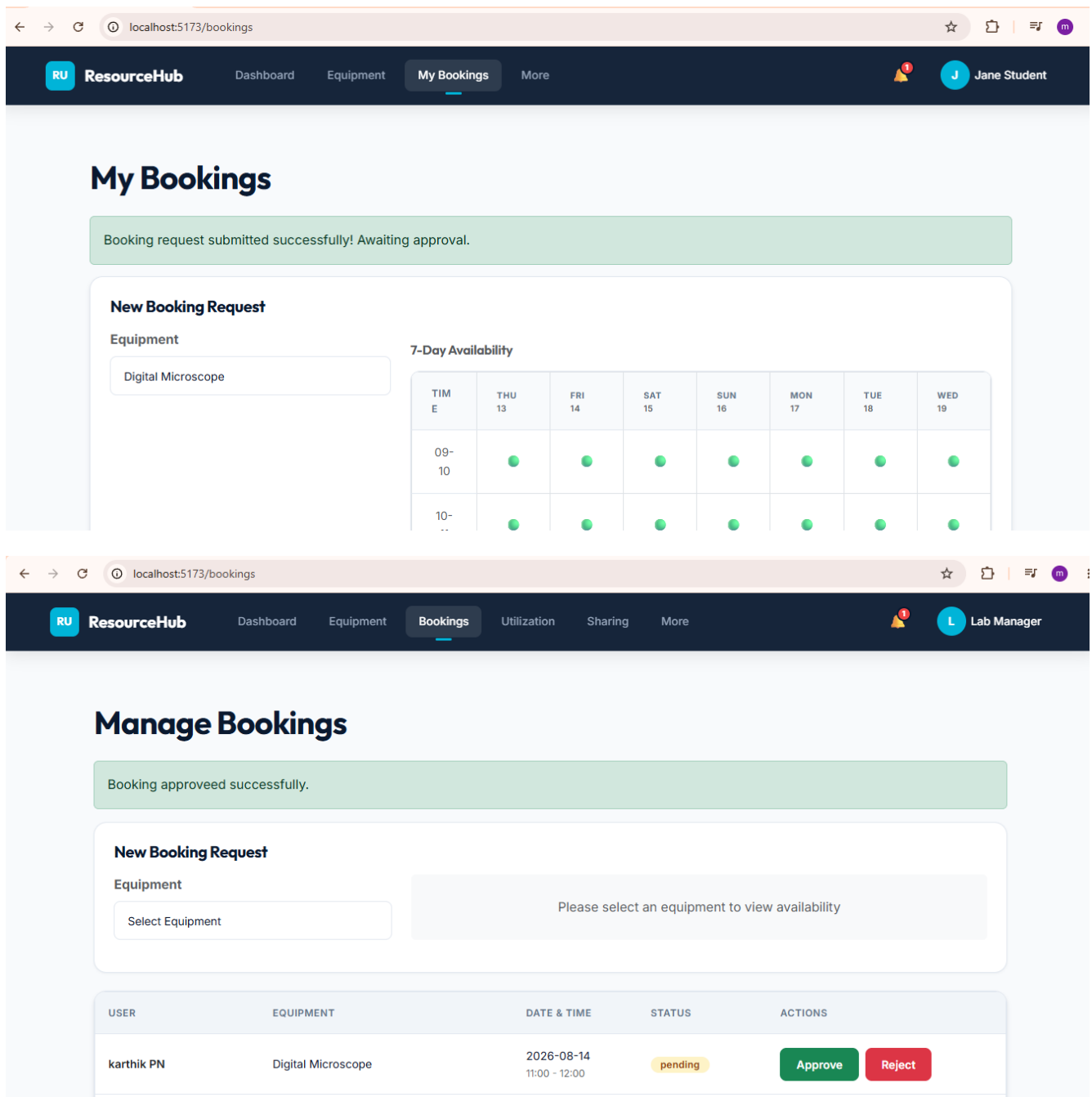


Figure 10.1: End-to-End Workflow Output

Screenshot to insert: Insert the most important final workflow/output screenshot here.

11. Frontend – API – Backend – Database Integration

The Milestone 2 implementation should demonstrate that the frontend is connected to backend APIs and database operations rather than relying only on static interface elements.

- Frontend sends user actions to the appropriate API.
- Backend validates the request and applies business rules.

- Database stores equipment, booking, utilization, sharing, and related data.
- The API response is returned to the frontend.
- The dashboard updates using the resulting data.

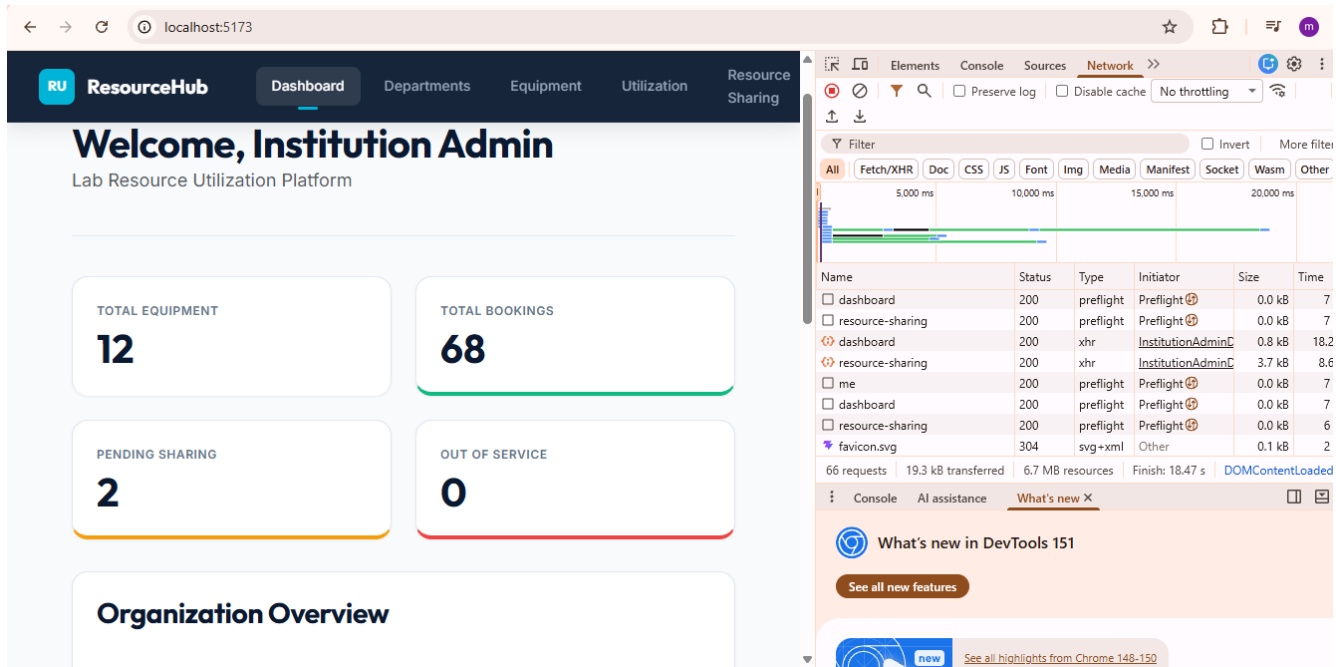


Figure 11.1: Frontend/API Integration

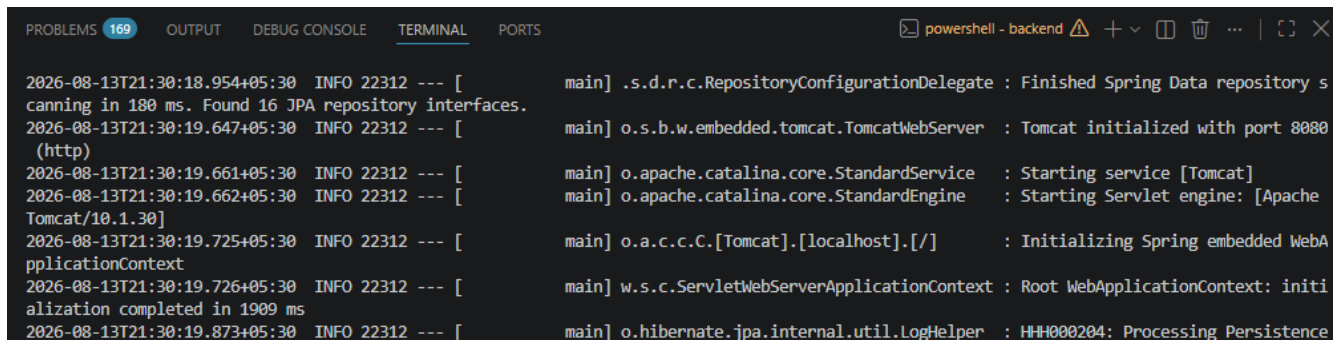


Figure 11.2: Backend / Database Output

Screenshot to insert: Insert backend console, API response, or database screenshot if available.

12. Equipment and Booking Statuses

Status Type	Statuses
Equipment Status	Available, Booked, Under Maintenance, Out of Service, Retired, and In Use where applicable
Booking Status	Pending Approval, Confirmed, In Use, Completed, Cancelled, No Show

13. Testing and Validation

The following validation points should be demonstrated using the implemented application and corresponding output screenshots.

Test Case	Expected Result
TC-01	Role-based login works correctly.
TC-02	Correct dashboard loads for each role.
TC-03	Role-based navigation is enforced.
TC-04	Unauthorized access is prevented.
TC-05	Dashboard data is connected to backend APIs/database.
TC-06	Major buttons perform actual operations.
TC-07	Researcher can check availability and book equipment.
TC-08	Waitlist functionality is connected to booking availability.
TC-09	Lab Manager / Department Head can view utilization and heatmap.
TC-10	Sharing requests can be viewed and approved/rejected.
TC-11	Institution Administrator can view organization-level information.
TC-12	Milestone 2 workflows operate as an integrated end-to-end flow.

14. Milestone 2 Results

Milestone 2 integrates user roles, equipment utilization, booking, waitlist, resource sharing, analytics, and dashboard functionality into a connected workflow. The final implementation should demonstrate actual operations through the frontend and its connected backend services.

The screenshots inserted in this document provide visual evidence of the implemented functionality. Each figure should be accompanied by a short explanation of the operation performed and the resulting output.

15. Conclusion

Milestone 2 establishes the core resource-utilization workflow of the Lab Resource Utilization Platform. The implementation focuses on monitoring equipment usage, identifying utilization and idle-time patterns, managing bookings and waitlists, supporting resource sharing, and providing role-specific dashboards and analytics. The completed functionality demonstrates how researchers, laboratory managers, department heads, and institution administrators can interact with shared laboratory resources according to their roles.

16. Screenshot Checklist

☐ Login / authentication screenshot

- ☐ Researcher / Student dashboard
- ☐ Equipment listing/search
- ☐ Equipment availability
- ☐ Booking form
- ☐ Successful booking
- ☐ Equipment status change
- ☐ Lab Manager dashboard
- ☐ Utilization percentage / statistics
- ☐ Utilization heatmap
- ☐ Idle equipment
- ☐ Resource sharing
- ☐ Sharing request
- ☐ Approval / rejection
- ☐ External booking/access
- ☐ Demand analytics
- ☐ High-demand equipment
- ☐ Waitlist option
- ☐ Waitlist status
- ☐ Double-booking prevention
- ☐ Department Head dashboard
- ☐ Institution Administrator dashboard
- ☐ API/backend/database evidence
- ☐ End-to-end final output